The table below lists the See3CAM_37CUGM VID, PID, and Supported OS.

Description	Specifications
Vendor ID	0x2560
Product ID	0xC13A
Supported OS	Windows 10 and Windows 11
	Ubuntu 18.04, Ubuntu 20.04 and Ubuntu 22.04

Table 1: See3CAM_37CUGM VID, PID, and Supported OS

3.1 Features

The features of See3CAM_37CUGM are as follows:

- 3.2MP IMX900 Monochrome image sensor with 1/3.1" optical form factor
- USB 3.2 Gen 1 device with Type-C reversible interface connector
- Standard M12 lens holder for use with customized optics or lenses for various applications
- Still capture is available in supported resolutions
- Video streaming available in supported resolutions
- Supported Controls
    - UVC Controls
        - Manual Gain
        - Exposure
    - HID Controls
        - Camera mode
        - Orientation mode
        - Image Burst
        - Strobe mode
        - Exposure mode
        - Manual Gain
        - Manual Exposure
        - Target Brightness
        - Exposure compensation
        - Anti-flicker
        - Read Statistics
        - Read Temperature
        - Fast Auto Exposure Feature
        - Self-Trigger feature
        - Restore default
        - Read Firmware Version
        - Read ISP Version
        - Save Configuration
- Operating temperature range: -30°C to +70°C
- Light weight, versatile, and portable design
- RoHS compliant

3.2 Image Sensor Features

The table below lists the key specifications of See3CAM_37CUGM.

Sensor Specification	
Type/Optical Size	1/3.1" Optical format Monochrome image sensor
Resolution	3.2MP
Pixel Size	2.25 µm
Sensor Active Array Size	2064 x 1552

Table 2: Image Sensor Features

3.3 Supported Resolutions and Frame Rates

The table below lists the various resolutions and frame rates of See3CAM_37CUGM.

Format	Resolution	Frame Rate		% Crop	
		USB 3.2 Gen 1	USB 2.0	Horizontal	Vertical
Y12	2064 x 1552	51	6	0	0
		30			
		25			
	1920 x 1080	72	10	6.97	30.14
		60			
		50			
	640 x 480	145	70	68.99	69.07
		120	60		
		100	50		
Y8	2064 x 1552	72	10	0	0
		60			
		50			
	1920 x 1080	100	15	6.97	30.14
		60			

	1024 x 768 (HDR*)	50	40	0	0
	640 x 480	198	100	68.99	69.07
		120	60		
		100			

Table 3: Supported Resolutions and Frame Rates

HDR: When the user switches to 1024x768 Resolution in Y8 Format, by default, the streaming will be in HDR mode. Refer to Appendix to know more about HDR.

4 Pin Description

The See3CAM_37CUGM camera has a USB 3.2 Gen 1 Type-C connector, which can interface with a PC. The below figure shows the location of the Type C connector (CN1) and GPIO header (CN4) connectors.

Figure 4: Location of Connector CN1 and CN4 on See3CAM_MIPI_BASE15_H01R1

The table below lists the pinouts of the USB 3.2 Gen 1 connector (CN1), which connects the See3CAM_37CUGM board to the PC through a USB cable. This is a standard USB 3.2 Gen 1 Type-C connector.

Pin No	Signal	Description	Pin No	Signal	Description
A1	GND	Ground	B12	GND	Ground
A2	SSTXP1	SuperSpeed differential pair1, TX, positive	B11	SSRXP1	SuperSpeed differential pair 2, RX, positive
A3	SSTXN1	SuperSpeed differential pair 1, TX, negative	B10	SSRXN1	SuperSpeed differential pair 2, RX, negative
A4	VBUS	Power	B9	VBUS	Power

A5	CC1	Configuration channel	B8	SBU2	-
A6	DP1	Hi-Speed differential pair, position 1, positive	B7	DN2	Hi-Speed differential pair. Position 2. negative
A7	DN1	Hi-Speed differential pair, position 1, negative	B6	DP2	Hi-Speed differential pair. Position 2. positive
A8	SBU1	-	B5	CC2	Configuration channel
A9	VBUS	Power	B4	VBUS	Power
A10	SSRXN2	SuperSpeed differential pair 4, RX, negative	B3	SSTXN2	SuperSpeed differential pair 3, TX, negative
A11	SSRXP2	SuperSpeed differential pair 4, RX, positive	B2	SSTXP2	SuperSpeed differential pair 3, TX, positive
A12	GND	Ground	B1	GND	Ground

Table 4: Pin Descriptions of CN1 Located on See3CAM_MIPI_BASE15_H01R1

The table below lists the pinouts of the GPIO Header connector (CN4).

CN4 Pin no	Signal Name	Pin Type	Description	Remarks
1	VCC5V	Power	5V Power	Output
2	VCC5V	Power	5V Power	Output
3	VCC5V	Power	5V Power	Output
4	GND	Power	Ground	-
5	GND	Power	Ground	-
6	CAM_HSYNC / FX3_I2C_SCL	Bidirectional (PU)	HSYNC from Sensor/ FX3 I2C SCL	Default connected to FX3 I2C (Pulled up to 1.8V)
7	CAM_VSYNC / FX3_I2C_SDA	Bidirectional (PU)	VSYNC from Sensor/ FX3 I2C SDA	Default connected to FX3 I2C (Pulled up to 1.8V)
8	EXT_TRIG_2	Input	Trigger 2	
9	CAM_FLASH	Output	Sensor exposure period monitor	Signal is Active low
10	EXT_TRIG_1	Input (PD)	Trigger 1	
11	GND	Power	Ground	
12	GND	Power	Ground	

Table 5: General Purpose Pin Description (CN4)

PU - Internally Pulled-up and PD - Internally Pulled-down

5 Electrical Specification

The below sections list the electrical specifications, recommended operating conditions, and power consumption details of See3CAM_37CUGM.

- Recommended Input Power
- Power Consumption Details
- Operational Temperature Range

The values described in this section are measured in the e-con Systems lab, and this can be used as a reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

5.1 Recommended Input Power

The table below lists the recommended input power of See3CAM_37CUGM.

Parameter	Typical Operating Voltage(V)	Typical Power Consumption (W)	Peak Consumption(W)
2064x1552 resolution at 51fps in Y12 format in USB 3.2 interface	5	1.42	2.38

Table 6: Recommended Input Power

5.2 Power Consumption Details

The tables below list the power consumption details of See3CAM_37CUGM for various resolutions and fixed frame rates in different modes.

Format	Resolution	Frame Rate (fps) USB 3.2 Gen	Voltage(V)	Current Consumption(mA)	Power Consumption(W)
Y12	2064 x 1552	51	5	284	1.42
		30	5	281	1.405
		25	5	279	1.395
	1920 x 1080	72	5	283	1.415
		60	5	287	1.435
		50	5	283	1.415
	640 x 480	145	5	269	1.345
		120	5	269	1.345
		100	5	267	1.335
Y8	2064 x 1552	72	5	266	1.33
		60	5	282	1.41
		50	5	283	1.415
	1920 x 1080	100	5	282	1.41
		60	5	284	1.42

	1024 x 768	50	5	264	1.32
	640 x 480	198	5	272	1.36
		120	5	271	1.355
		100	5	284	1.42

Table 7: Power Consumption Details in USB 3.2

Format	Resolution	Frame Rate (fps) USB 2.0	Voltage(V)	Current Consumption(mA)	Power Consumption(W)
Y12	2064 x 1552	6	5	253	1.265
	1920 x 1080	10	5	257	1.285
	640 x 480	70	5	260	1.3
		60	5	260	1.3
		50	5	256	1.28
Y8	2064 x 1552	10	5	278	1.39
	1920 x 1080	15	5	277	1.385
	1024 x 768	40	5	257.8	1.289
	640 x 480	100	5	275	1.375
		60	5	276	1.38

Table 8: Power Consumption Details in USB 2.0

5.3 Operational Temperature Range

The table below lists the operational temperature range of See3CAM_37CUGM.

Temperature Range	Parameter Description
-30°C to +70°C	operational temperature

Table 9: Operational Temperature Range

Note: You can choose different lenses as per your requirements. You are advised to make the necessary arrangements for the products to dissipate the heat generated in the module and maintain the operating temperature below 70°C.

6 Mechanical Specifications

See3CAM_37CUGM is a two-board solution, as mentioned below:

- Camera module board (e-CAM315_CUMI900M_MOD_H02R1)
- Adapter board (SEE3CAM_MIPI_BASE15_H01R1)

The below figures show the mechanical dimensions of the boards.

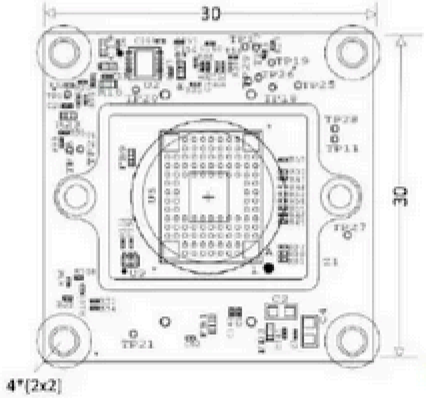

Figure 5: e-CAM315_ CUMI900M MOD H02R1 – Top View Mechanical Dimensions

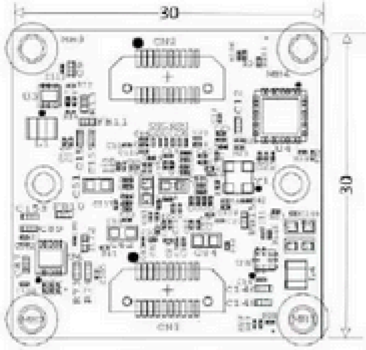

Figure 6: e-CAM315_CUMI900M_MOD_H02R1 – Bottom View Mechanical Dimensions

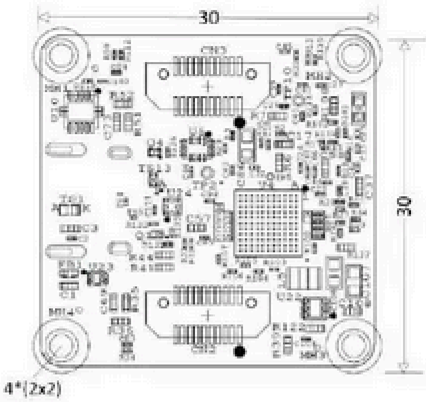

Figure 7: See3CAM_MIPI_BASE15_H01R1 Top view Mechanical Dimensions

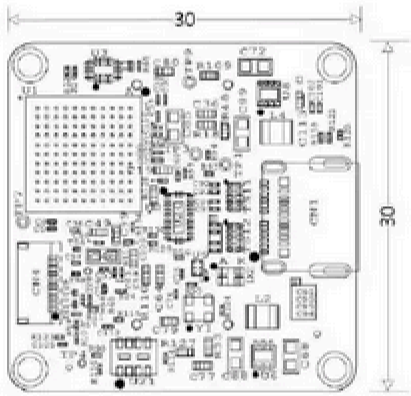

Figure 8: See3CAM_MIPI_BASE15_H01R1 Bottom View

Note: All dimensions are in mm.

6.1 Lens Holder Dimensions

The below figure shows the dimension details of the S-Mount metal lens holder.

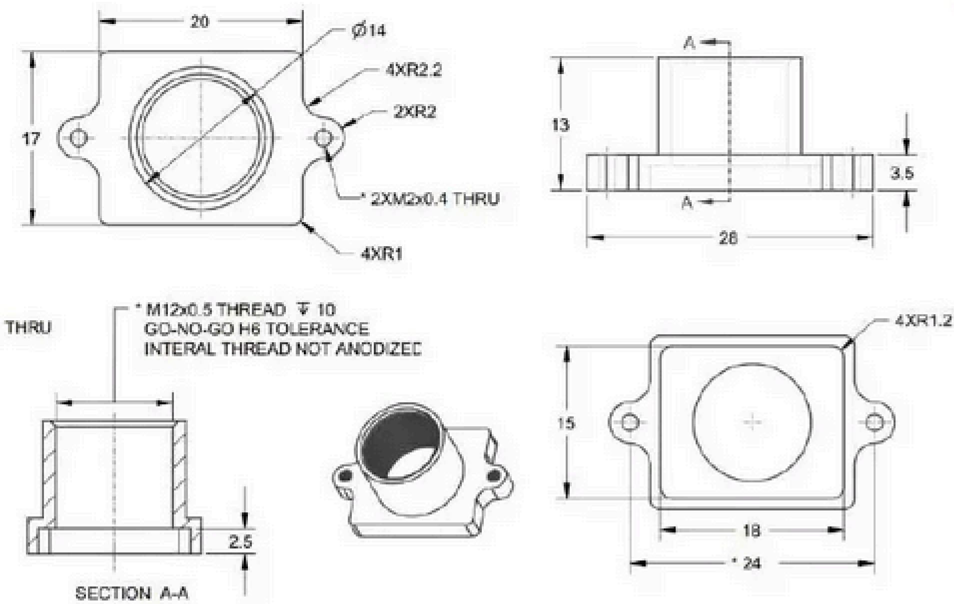

Figure 9: S-Mount Metal Lens Holder Outline Dimension

Note: All dimensions are in mm.

7 Mechanical Part Details

The table below lists the mechanical accessories required for See3CAM_37CUGM.

Part	Quantity	Specification
Lens holder	1	Lens Holder M12 x 0.5 Thread Portion, Aluminum Height 9mm, OD 14mm, Hole Pitch 20 mm, BFL (Back Focal Length) 2.50mm
Lens holder screw	2	Screw M2 machine screw stainless steel head diameter 4mm max, thread diameter 2mm max, length 5mm
USB cable	1	USB 10Gbps (USB 3.1, USB 3.x Gen 2, Superspeed+) Cable A Male to C Male 3.28' (1.00m) Shielded
Screw	4	Screws M2 Phillips Pan Head Stainless Steel Head Dia-4.0mm, Thread Dia-2mm, Length-12mm
Spacer	4	Metal Spacer Round Unthreaded Length 6mm, OD 3.8mm, ID 2.2mm
Nut	4	Nut M2x0.4 Hex Stainless steel Width Across Flats 4mm max, Width across corners 4.32mm min, Thickness 1.6mm max

Table 10: Mechanical Part Details

8 Orientation of See3CAM_37CUGM

The figure below shows the image orientation of See3CAM_37CUGM on the e-cam viewer with respect to the USB connector.

Figure 10: Image Orientation with respect to USB Connector

9 Application Notes

9.1 Y12 Pixel Format

In See3CAM_37CUGM, sensor data will be transferred to the host controller through the Parallel interface. The transmission of 12-bit raw data is accomplished by packing the 12-bit pixel data to look like 8-bit data format.

The below table describes the number of pixels and bytes constraints.

Pixels	Bytes	Used Bits
2	3	24

Table 11: Number of Pixels and Bytes Constraints

In Y12 format, every consecutive two pixels are packed in three bytes as follows:

Consider two pixels, A and B,

A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	B_{11}	B_{10}	B_9	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	A_3	A_2	A_1	A_0

Byte 1 **Byte 2** **Byte 3**

So, to get the first pixel (A) data, the least significant nibble of the third byte should be appended with the first byte as below,

A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0

Similarly, the second pixel (B) data can be obtained by appending the most significant nibble of the third byte with the second byte as follows:

B_{11}	B_{10}	B_9	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0

Example:

0x49	0x0A	0xA8

Byte-1 **Byte-2** **Byte-3**

Pixel 1 – 0x498

Pixel 2 – 0x0AA

This procedure must be followed to obtain the pixel data in Y12 format if the user writes their own application.

9.2 Trigger – Exposure Control

The See3CAM_37CUGM offers the ability to synchronize the start of the image sensor's exposure with this triggering action. This synchronization is controlled on the image sensor using one trigger signal. Additionally, the camera offers the flexibility to program the exposure time.

In this exposure control trigger mode, the user needs to provide an external trigger signal through pin 10 of the GPIO connector (CN4) to start the frame from the sensor. The low pulse width of the external trigger signal decides the exposure time of the frame. In the figure below, a low pulse width of 15.6ms is given as the input trigger, so the output frame exposure will be 15.6ms.

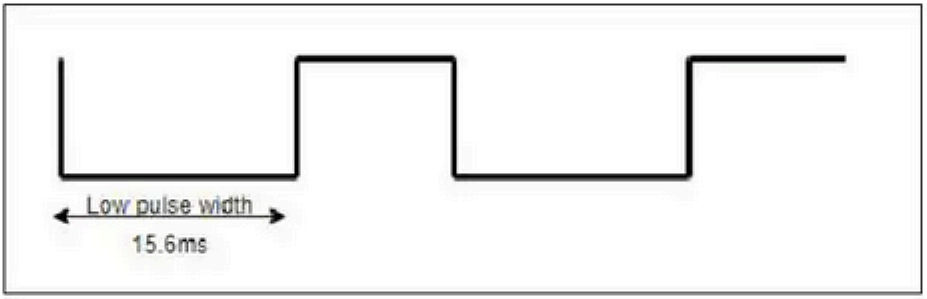

Figure 11: Exposure Control – Trigger Pulse Timing Diagram

Note:

The external trigger input must follow the below conditions.

- Input trigger period <=streaming resolution maximum FPS period.
- One cycle period = 1/(Number of frames per second).
- T_{HIGH} = One cycle period – T_{LOW}.
- T_{LOW} = Required exposure time is given by the user.

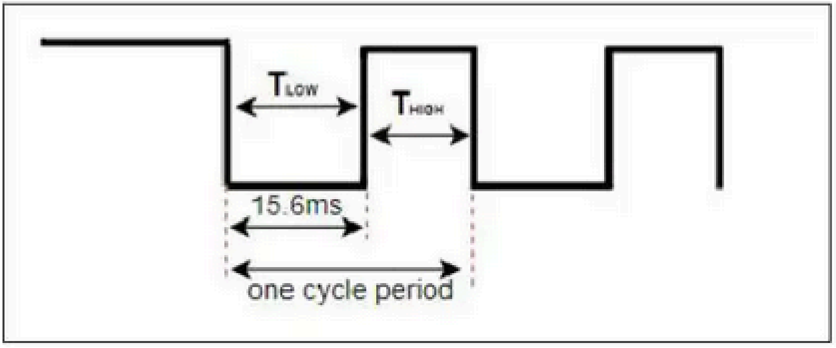

Figure 12: Timing Diagram for Exposure Trigger

9.3 Strobe Output

Strobe output can be obtained in the Pin 9 of the GPIO connector (CN4). The exposure time of the current frame is obtained by using this pin 9. The negative pulse width of the signal indicates the exposure time of the current frame. By default, the strobe mode will be OFF.

Example: The strobe timing diagram for default exposure value 15.6ms.

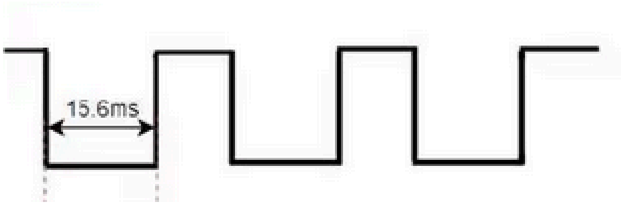

Figure 13: Strobe Timing Diagram

9.4 Fast-Auto Exposure Feature

Fast auto exposure is a specialized mode of the See3CAM_37CUGM device in which the sensor automatically adjusts exposure in real-time, ensuring that every frame is well-lit and correctly exposed, even in fast-changing light conditions. By default, the Fast Auto Exposure mode will be disabled.

Auto Exposure Execution frames:
The Auto Exposure Execution Frames indicate the number of frames considered for calculating Auto Exposure statistics. It ranges from 6 frames to 1 frame. The default value is 6.

Note:

Increasing the number of AE execution frames speeds up the Auto Exposure settling rate but may result in frame drops. Conversely, decreasing the AE execution frames slows down the Auto Exposure settling rate but preserves the frame rate.

No of AE Execution Frames	Frame rate achieved when FAE enabled in Y12 Format (fps)	Frame rate achieved when FAE enabled in Y8 Format (fps)
6	17.97	24.65
5	18.94	26.02
4	20.01	27.64
3	21.22	29.42
2	22.58	31.46

1	24.13	33.79

Table 12: Number of AE Execution frames and corresponding framerate in Y12 and Y8 format

FAE Target Brightness:

Target brightness act as the reference luminance level, guiding the sensor's AE algorithm to adjust the auto exposure settings to maintain a consistent brightness level across different lighting conditions.

Format	Minimum Value	Maximum Value	Default Value
Y12	0	4095	409
Y8	0	255	50

Table 13: FAE Target Brightness Minimum, Maximum and default value in Y12 and Y8 format

Restrictions on Fast Auto Exposure:

- Cross still format and cross still resolution support is restricted for fast auto exposure. User should ensure that still capture format and resolution is as same as the video capture format and resolution.
- Fast auto exposure mode will be disabled, If the user switches from 2064x1552 resolution to another resolution with fast auto exposure mode enabled.
- Fast auto exposure mode is restricted in USB 2.0.

9.5 Self-Trigger Feature

The self-trigger feature can designate two areas from the effective pixels: the sensing area and the capturing area. When the number of pixels surpasses the level and count threshold in the sensing area, the sensor triggers capturing an image in the capturing area.

Sensing Area and Capturing Area Selection

The capturing area is restricted to (H x V) 2064 x 1552 pixels. Thus, the user will get streaming of the capturing frame (2064 x 1552) only when both the threshold conditions of sensing frames are met.

The user has the flexibility to change only the sensing area size (H x V) and the position of the sensing area size in the e-cam View window. The minimum and maximum values of the horizontal area size of the sensing frame are 640 and 2064 pixels, respectively. The minimum and maximum value of the vertical area of the sensing area size will be 480 and 1552 pixels, respectively.

Note: The sensing area and cropping area can be overlapped.

To choose the position of the sensing area in the e-CAMView window, the user must choose/fill out all the required values in self-trigger mode and set them using the available set button in the Self-Trigger tab. After clicking the Set Button, close the extension unit, and locate the position of the sensing frame by clicking left mouse button in the e-CAMView window. Thus, the sensing area size specified by the user will be applied around the coordinate located by the user. The default value of Horizontal sensing area size and vertical sensing area size will be 2064 and 1552 respectively, where the default position of the sensing area will be (0,0) thus the sensing frame covering the entire pixel area.

Threshold H and L Values

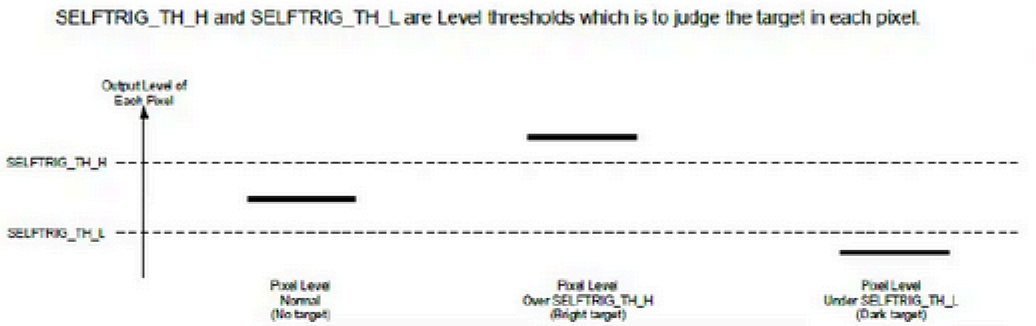

Figure 14: Self-Trigger Level Threshold H and L Side

The level threshold can be set to a high side (bright target) and a low side (dark target). When the pixel level is above the user-specified Level threshold H value, then the pixel is considered a bright pixel, and when the pixel level is below the user-specified Level threshold L value, then the pixel is considered a dark pixel. The default value for Level threshold H side is 4095 and Level threshold L side is 0.

Format	Level Threshold H (Min)	Level Threshold H (Max)	Level Threshold L (Min)	Level Threshold L (Max)
Y12	0	4095	0	4095
Y8	0	255	0	255

Table 14: Self Trigger – Level threshold H, L side - Minimum and Maximum Values

Count Threshold H and L Values

The user-specified values for Count Threshold H and Count Threshold L are the threshold of the number of pixels that are over the Level threshold H and under the Level threshold L in the sensing area, respectively, which judge whether to capture the image or not. In case over one of the two count thresholds, the trigger is generated internally, and the capturing area is captured, which is available in the streaming.

Note: All pixels in the sensing area aren't sensing.

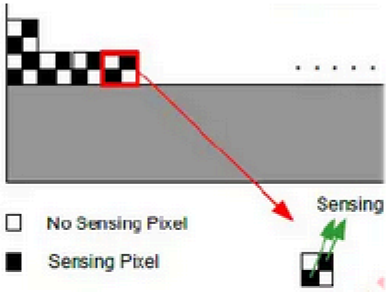

Figure 15: Self Trigger – Sensing Pixels of Sensing Area

For example, when the user specifies sensing area pixels as 2064 x 1552, only 16,01,664 ((2064x1552)/2) number of pixels will be considered for Count Threshold H and L Values.

Format	Level Threshold H (Min)	Level Threshold H (Max)	Level Threshold L (Min)	Level Threshold L (Max)
Y12	0	16,01,664	0	16,01,664
Y8	0	16,01,664	0	16,01,664

Table 15: Self Trigger – Count Threshold H, L Side - Minimum and Maximum Values

Since the maximum value of the count threshold H and L depends on the sensing area, the user should select their appropriate count threshold H and L values according to the sensing area.

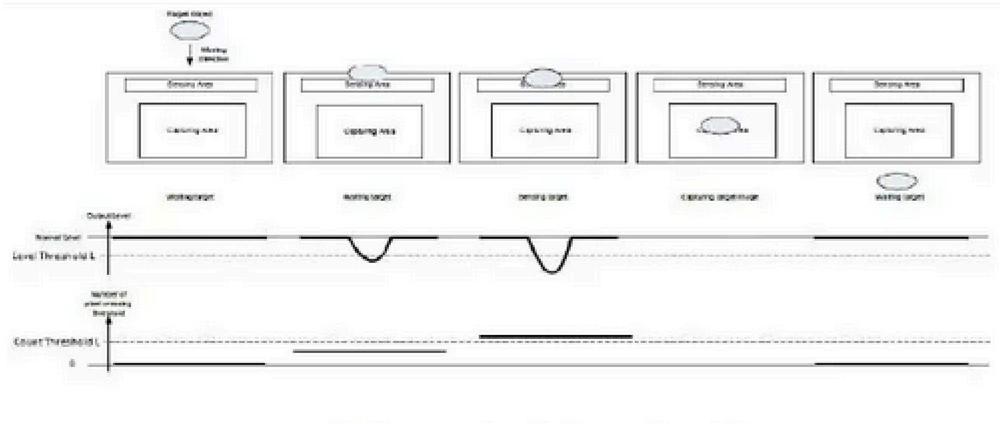

Figure 16: Self Trigger – Low-level threshold

The above image represents the function of self-trigger (capturing the target object in the capturing area when the number of pixels in the sensing area which are below the level threshold L is more than the count of count threshold L side).

Sensing Area and Capturing Area Gain:

Users can set the sensing area gain and capturing area gain separately using the respective gain sliders. The minimum and maximum values of both sensing and capturing area gain are 0dB and 24dB, respectively. The default values are 0dB for both sensing and capturing area gain.

Restrictions on Self Trigger:

Streaming:

When the threshold conditions of the sensing frame are not met, no trigger is generated internally, thus resulting in 0 frames per second in e-CamView. Once the sensing frame threshold conditions are satisfied, the capturing area frame will be received, and streaming will continue until the sensing frame threshold conditions are met.

Exposure:

In self-trigger mode, the exposure of both the sensing area and the capturing area is dependent on the Sensing area's vertical size. If the user chooses 2064x1552 (H x V) as the sensing area size, the exposure applied in both the sensing and capturing area will be 18.89ms in Y12 and 13.29ms in Y8. But if the user chooses a smaller sensing area size, i.e., 640x480 (H x V), then the exposure applied in both the sensing and capturing area will be 6.46ms in Y12 and 4.5ms in Y8.

The tables below list the exposure of the capturing and sensing area based on the vertical sensing area.

Sensing Area Vertical	Exposure
	Y12
480	6.46ms
528	7ms
616	8ms
704	9ms
792	10ms
872	11ms
960	12ms
1048	13ms
1136	14ms
1224	15ms
1304	16ms

1392	17ms
1480	18ms
1552	18.89ms

Table 16: Exposure of Capturing and Sensing Area based on Vertical Sensing Area in Y12 Format

Sensing Area Vertical	Exposure
	Y8
480	4.5ms
536	5ms
656	6ms
776	7ms
904	8ms
1024	9ms
1152	10ms
1272	11ms
1400	12ms
1520	13ms
1552	13.29ms

Table 17: Exposure of Capturing and Sensing Area based on Vertical Sensing Area in Y8 Format

Frame Rate

In self-trigger, the user cannot expect a fixed frame rate since the trigger is based on threshold conditions satisfied in the sensing area.

Sensing Frame Output

The user cannot get the image of the sensing area. Only the capturing area image will be received as output in the streaming when the sensing area threshold conditions are met.

Still Capture Support

In QtCAM, Still Capture option is restricted for self-trigger.
In e-CAMView, Cross still format and cross still resolution support is restricted for self-trigger. User should ensure that still capture format and resolution is as same as the video capture format and resolution.

USB 2.0

Self-trigger mode is restricted in USB 2.0.

9.6 HDR Mode

In See3CAM_37CUGM, by default the 1024 x 768 resolution of Y8 format will be streaming in HDR mode. No other resolution of Y8 and Y12 will support HDR.

Auto Exposure HDR

Auto Exposure HDR refers to the High Dynamic Range imaging mode where the exposure settings are automatically adjusted by the See3CAM_37CUGM camera's AE algorithm.

For HDR processing in auto exposure mode, See3CAM_37CUGM AE algorithm applies 3 Exposure values to each of the individual 2-by-2 pixels units as mentioned in the diagram. Each 2-by-2 pixel units of the entire 2064 x 1552 pixel area is treated as a group and 3 types of exposure values is applied to it for HDR processing(Long Exposure value, Middle Exposure Value, short Exposure Value).

Figure 16 : 2-by-2 Pixel Units of the pixel array

See3CAM_37CUGM's Auto Exposure algorithm will determine the auto exposure statistics from the previous middle exposure(ME) frame and corresponding the medium exposure (ME), long exposure (LE), short exposure (SE) are applied based on the table below.

For example.,
when See3CAM_37CUGM camera is exposed to the bright light condition, the AE algorithm will determine the exposure setting (Middle Exposure value in HDR) as 120us.Based on the table given below, the Middle exposure value falls in the range between 72us <120us <192us. Thus, applied short exposure will be (120us/24) = 5us (Ratio of Medium exposure to short exposure is 24) and the applied long exposure will be (120us*24) = 2880us (Ratio of Long exposure to medium exposure is 24).

S. No	Short Exposure (µs)	Medium exposure to Short Exposure Ratio (x)	Medium Exposure (us)	Long Exposure to Medium Exposure Ratio (x)	Long Exposure (us)
1	3	24	72	24	1728
2	8	24	192	16	3072

3	15	24	360	12	4320
4	25	24	600	8	4800
5	41	24	984	6	5904
6	72	24	1728	4	6912
7	122	24	2928	3	8784
8	208	24	4992	2	9984
9	351	24	8424	1.5	12636
10	600	24	>14400	1	14400

Table 18:Auto Exposure HDR – SE, LE Exposure value calculation based on ME

Note:

The user can read the applied middle exposure value in AE algorithm in the **Read statistics** text box of Extension unit control. The Read statistics text box will display the middle exposure value of HDR processing, only in the 1024 x 768 resolution of Y8 format that too with auto exposure mode enabled.

Manual Exposure HDR

Manual Exposure HDR refers to the High Dynamic Range Imaging mode where the user manually sets the middle exposure value and based on the user configured middle exposure value, the short exposure and long exposure value are automatically calculated based on *Table 18* for HDR processing.

Note

The User should manually provide the middle exposure in the **Manual Exposure text box** of extension unit control where the middle exposure value should be a multiple of 24. The Manual Exposure text box will accept the middle exposure value of HDR processing, only in the 1024 x 768 resolution of Y8 format that too with manual exposure mode enabled.

Limitations of HDR

(i) Cross still format or Cross still resolution is not supported in HDR. The user should initially configure the Y8 format and 1024 x 768 resolution in still capture pin and then should proceed for still capture.

(ii) In low-light conditions, white pixel noise becomes more noticeable. A workaround is being implemented to optimize this noise. Therefore, it is recommended to use HDR in well-lit environments.

Revision History

Rev	Date	Description	Author
1.0	04-DEC-2024	Initial draft	Camera Products Team
1.1	11-FEB-2025	ISP Auto Exposure Features Enabled	Camera Products Team
1.2	14-MAR-2025	HDR, FAE Controls added, modified frame rate	Camera Products Team
1.3	18-MAR-2025	FAE, Self trigger Still capture support added	Camera Products Team